Name ; S. Hari Rebavan

Reg no : 19251172.

Slot : D

Course code : ITA1402.

23/25

SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ITA0506-COMPUTER VISION

MCQ TEST

1. At a particular pixel, Ix = 4 and Iy = 3. What are the values of Ix^2 and Iy^2?
A) 8, 6 B) 16, 9 C) 12, 7 D) 4, 3

2. At a particular pixel, Ix = 5 and Iy = 2. What is the value of Ix × Iy?
A) 7 B) 10 C) 25 D) 4

3. For a window with Ix values [1, 2, 3], what is the value of $Sum(Ix^2)$?
A) 6 B) 9 C) 14 D) 36

4.Given the Harris matrix M = [10 4; 4 8], what is the determinant of M?
A) 64 B) 72 C) 80 D) 96

5. Given the Harris matrix M = [12 3; 3 8], what is the trace of M?
A) 5 B) 12 C) 20 D) 24

6. What is the main benefit of image enhancement?
A) To make important image features easier to identify
B) To remove all image information
C) To increase image distortion
D) To reduce the number of objects in an image

7. An 8-bit grayscale image can represent how many different intensity levels?
A) 8 B) 128 C) 256 D) 512

8. Which of the following best represents a grayscale image?
A) Each pixel contains only red, green, and blue components
B) Each pixel represents an intensity value
C) Each pixel contains only text information
D) Each pixel represents a geometric shape

9.If Sum(Ix^2) = 20, Sum(Iy^2) = 15, and Sum($Ix \times Iy$) = 5, which is the correct Harris matrix?

A)

$$\begin{bmatrix} 25 & 16 \\ 4 & 4 \end{bmatrix}$$

B)

$$\begin{bmatrix} 25 & 4 \\ 4 & 16 \end{bmatrix}$$

C)

$$\begin{bmatrix} 16 & 4 \\ 4 & 25 \end{bmatrix}$$

D)

$$\begin{bmatrix} 4 & 25 \\ 16 & 4 \end{bmatrix}$$

10. If determinant(M) = 100, trace(M) = 20, and k = 0.04, what is the Harris response R?

A) 60 B) 70 C) 80 D) 84

11. If Sum(Ix^2) = 25, Sum(Iy^2) = 16, Sum($Ix \times Iy$) = 4, and k = 0.04, what is the approximate Harris response R?

A) 180.00 B) 220.00 C) 277.24 D) 316.76

12. For an image point (x, y) = (4, 3), calculate rho using $\rho = x\cos(\Theta) + y\sin(\Theta)$ when Θ = 0 degrees.

A) 0 B) 3 C) 4 D) 7

13. For an image point (x, y) = (2, 5), calculate rho when theta = 90 degrees.

A) 2 B) 5 C) 7 D) 10

14. Three edge points belonging to the same straight line vote for the same parameter pair (rho, theta). What will be the accumulator value at that location?

A) 0 B) 1 C) 2 D) 3

15. A Hough accumulator contains the following voting values: A = 3, B = 12, C = 5, D = 2. If the minimum threshold for detecting a line is 5 votes, which parameter combinations represent possible lines?

A) A only B) B only C) B and C D) A, B and C

16.A straight line in an image contains some missing edge pixels. The Hough Transform is still able to detect the line because:

A) It requires every pixel of the line

B) The available edge pixels vote for common line parameters

) It automatically fills all missing pixels

) It removes the missing pixels before voting

17. For a point (x, y) = (5, 4), the Hough Transform is evaluated at theta = 90 degrees. What is the value of rho?

A) 4 B) 5 C) 9 D) 20

18. A Hough accumulator has the following values: 2, 4, 15, 3, and 6. Which value indicates the strongest candidate for a straight line?

A) 2 B) 4 C) 6 D) 15

19. Four edge points produce the following votes for three parameter combinations: Parameter A = 2 votes, Parameter B = 8 votes, Parameter C = 5 votes. If the detection threshold is 5 votes, how many lines are detected?

A) 1 B) 2 C) 3 D) 0

20. Which of the following is the main purpose of image preprocessing?
A) To improve image quality before further processing
B) To increase image noise
C) To reduce the image information
D) To convert an image into a video

21. Which of the following is a common image preprocessing technique used to reduce noise?
A) Smoothing B) Hough Transform C) Corner Detection D) Edge Linking

22. Which type of image processing operates directly on the pixel values of an image?
A) Spatial domain processing
B) Frequency domain processing
C) Feature matching
D) Object recognition

23. What is an edge in a digital image?
A) A region with constant intensity
B) A location where there is a significant change in intensity
C) A completely uniform region
D) A region containing only noise

24. Which technique is specifically designed to identify points where intensity changes significantly in multiple directions?
A) Smoothing B) Harris Corner Detection C) Histogram Processing
D) Image Compression

25. Why are corners considered important features in image processing?
A) They provide distinctive points that can be used for feature detection and matching
B) They always have the same intensity
C) They occur only in binary images
D) They are used only for noise removal